8 (a)(i)

$$F = \frac{1}{4\pi\varepsilon_0}\left(\frac{1.60\times10^{-19}\times1.60\times10^{-19}}{\left(10^{-14}\right)^2}\right)$$ C1

$$= 2.3\text{ N}$$ A1

(a)(ii)

$$\langle KE\rangle = \frac{3}{2}KT$$

$$70\times10^{3}\left(1.60\times10^{-19}\right) = \frac{3}{2}\left(1.38\times10^{-23}\right)T$$ C1

$$T = 5.4\times10^{8}\text{ K}$$ A1

(a)(iii) Some nuclei will be travelling faster / have greater kinetic energy to overcome electrostatic repulsion and hence cause fusion. B1

(a)(iv)

$$E = \Delta mc^2$$

$$18\times10^{6}\left(1.60\times10^{-19}\right) = \Delta m\left(3.00\times10^{8}\right)^2$$ C1

$$\Delta m = 3.2\times10^{-29}\text{ kg}$$ A1

(b)(i) graph: smooth curve in correct direction starting at (0,0) M1

D at $2t_{½}$ is 1.5 times that at $t_{½}$ (± 2 mm) A1

(b)(ii) *D* = number of parent nuclei that decayed

=original number of parent nuclei – number of parent nuclei remaining at time *t* M1

$= N_0 - N$

$= N_0 - N_0\,e^{-\lambda t}$

$= N_0\,(1 - e^{-\lambda t})$

(b)(iii) 1.

$${}^{238}_{92}\text{U} \rightarrow {}^{206}_{82}\text{Pb} + \text{X}\,{}^{4}_{2}He + \text{Y}\,{}^{0}_{-1}e$$

238 = 206 +4X $\Rightarrow$ X = 8 $\therefore$ 8 alpha decays M1

92 = 82 + 2X – Y

92 = 82 + 2(8) – Y M1

Y = 6 $\therefore$ 6 beta decays

(b)(iii) 2.

$$D = N_0\left(1 - e^{-\lambda t}\right)$$

$$\frac{D}{N_0} = 1 - e^{-\lambda t}$$

$$0.39 = e^{-\frac{\ln 2}{4.47\times10^{9}}t}$$ C1

$$t = 3.19\times10^{9}\text{ years}$$ A1

OR

$$N = N_0 e^{-\lambda t}$$

$$\frac{N}{N_0} = e^{-\frac{\ln 2}{t_{1/2}}t}$$

$$1 - 0.39 = e^{-\frac{\ln 2}{4.47\times10^{9}}t}$$

$$t = 3.19\times10^{9}\text{ years}$$

(b)(iv) 1.

$$N = N_0 e^{-\lambda t}$$

$$N_0 = \frac{N}{e^{-\lambda t}}$$ M1

$$D = N_0 - N$$ M1

$$= \frac{N}{e^{-\lambda t}} - N$$

$$= N\left(\frac{1}{e^{-\lambda t}} - 1\right)$$

(b) (iv) 2.

$$D = N\left(\frac{1}{e^{-\lambda t}} - 1\right)$$

$$\frac{D}{N} = \left(\frac{1}{e^{-\frac{\ln 2}{\lambda}t}} - 1\right)$$ C1

$$22.8 = \left(\frac{1}{e^{-\frac{\ln 2}{7.0\times 10^{8}}t}} - 1\right)$$

$$t = 3.17 \times 10^{9} \text{ years}$$ A1

(b) (iv) 3.

Any two from: B1

- Allows the mean to be calculated B1
- Some daughter product may have left the sample
- (Use of three series) allows identification of anomalous results/series
- Spread of results indicates uncertainty